Hybrid Machine Learning Models Using Soft Voting Classifier for Financial Distress Prediction

Minh Nguyen (Michigan State University)
Kien Cao-Van (Nguyen Tat Thanh University)
Le Gia Minh (Nguyen Tat Thanh University)
Tung X. Bui (University of Hawaii at Manoa)
Sukhwa Hong (University of Hawaii at Hilo)

The 8th Econometrics Conference of Vietnam
(ECONVN 2025)
January 2025

Outline

- Introduction
- Related work
- Methods
- Data
- Results
- Conclusion

Introduction

- Single machine learning models have been employed to predict financial distress in recent years.
- Predictive performances of financial distress prediction using single machine learning models can be relatively poor in the context of emerging economies (Nguyen et al., 2024).
- The question is how to improve the performances of the predictive models for given data.
- One solution is to build hybrid machine learning models.
- One reason is that multiple single machine learning algorithms can complement and augment each other to improve the overall performance when combined together.

Related work

The Journal of FINANCE

VOL. XXIII SEPTEMBER 1968 NO. 4

FINANCIAL RATIOS, DISCRIMINANT ANALYSIS AND THE PREDICTION OF CORPORATE BANKRUPTCY

EDWARD I. ALTMAN*

ACADEMICIANS SEEM to be moving toward the elimination of ratio analysis as an analytical technique in assessing the performance of the business enterprise. Theorists downgrade arbitrary rules of thumb, such as company ratio comparisons, widely used by practitioners. Since attacks on the relevance of ratio analysis emanate from many esteemed members of the scholarly world, does this mean that ratio analysis is limited to the world of "nuts and bolts"? Or, has the significance of such an approach been unattractively garbed and therefore unfairly handicapped? Can we bridge the gap, rather than sever the link, between traditional ratio "analysis" and the more rigorous statistical techniques which have become popular among academicians in recent years?

Figure 1: Altman (1968)

Related work

Journal of Accounting Research
Vol. 18 No. 1 Spring 1980
Printed in U.S.A.

Financial Ratios and the Probabilistic Prediction of Bankruptcy

JAMES A. OHLSON*

Figure 2: Ohlson (1980)

Related work

- Combine different types of methods
  - PCA-ANN (Adisa et al., 2019).
  - CSFNN model: Cuckoo search and feedforward neural network (Marso and El Merouani, 2020).
- Combine different machine learning models
  - Deep belief network and support vector machines (Lanbouri and Achchab, 2015; Huang and Yen, 2019).

Methods

Six single Models

- Logistic regression (LR)
- Support vector machine (SVM)
- Decision trees (DT)
- Random forests (RF)
- Extreme gradient boosting (XGBoost or XGB)
- Light gradient boosting machine (LightGBM or LGBM).

Methods

Hybrid Models Using Soft Voting Classifier:

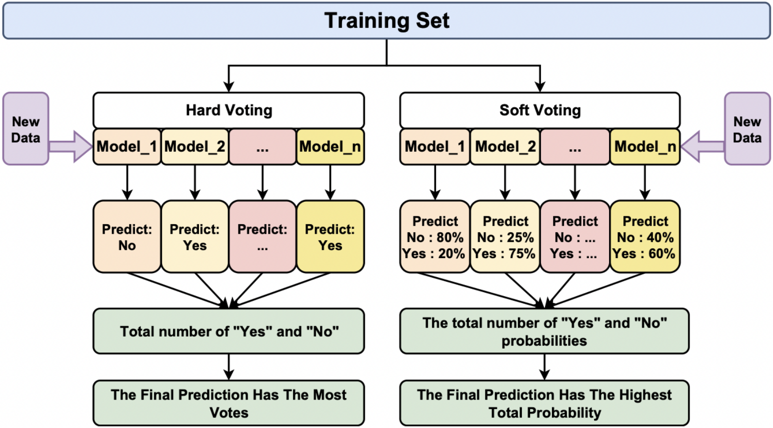

Figure 3: Comparison the training mechanisms of soft vs. hard voting

Source: Cao-Van et al. (2024)

Methods

Type	n
Hybrid-2 models	15
Hybrid-3 models	20
Hybrid-4 models	15
Hybrid-5 models	5
Hybrid-6 model	1
Total hybrid models	56
Total single and hybrid models	62

Table 1: Number of hybrid models

Data

- Data sources
  - Refinitiv Eikon database
  - Ho Chi Minh City Stock Exchange
  - Hanoi Stock Exchange
- Period: 2011-2021.
- Imbalanced data: minority vs. majority class.

Data

Table 2: Number of observations before and after SMOTE

	Train data	Percent	Test data	Percent
		Before SMOTE		
Delisted firms	414	4.08%	117	4.61%
Non-delisted firms	9,734	95.92%	2,420	95.39%
All firms	10,148	100%	2,537	100%
		After SMOTE		
Delisted firms	9,734	50%	117	4.61%
Non-delisted firms	9,734	50%	2,420	95.39%
All firms	19,468	100%	2,537	100%

Source: Nguyen et al. (2024)

Variables

Variable	Type	Definition
wcapat	Liquidity	Working Capital/Total Assets
reat	Profitability	Retained Earnings/Total Assets
ebitat	Profitability	Earnings Before Interest and Taxes/Total Assets
mvdebt	Solvency	Market Value of Equity/Total debt
saleat	Management's capability	Sales/Total Assets
size	Size	log(total assets/GNP price-level index)
ltat	Solvency	Total liabilities/ Total assets
lctact	Solvency	Total current liabilities/ Total current assets
OENEG	Solvency	Dummy: Total liabilities exceed total assets
roa	Profitability	Net income/Total assets
cfodebt	Solvency	Funds provided by operations/ Total liabilities
INTWO	Solvency	Net income <0 for in the last 2 years
CHIN	Management's capability	$\frac{\text{Net income}(t)-\text{Net income}(t-1)}{\lvert\text{Net income}(t)\rvert+\lvert\text{Net income}(t-1)\rvert}$

Table 3: List of Altman's and Ohlson's variables

Results

	Single model	Altman	Ohlson	All
Accuracy	LR	0.902	0.875	0.901
	SVM	0.902	0.892	0.898
	RF	0.851	0.885	0.861
	DT	**0.939**	0.918	0.940
	XGB	**0.939**	**0.934**	**0.962**
	LGBM	0.925	0.926	0.958
Balanced accuracy	LR	0.794	0.808	0.879
	SVM	0.859	0.813	0.877
	DT	0.826	0.786	0.802
	RF	0.812	0.826	0.866
	XGB	0.846	0.819	0.883
	LGBM	**0.863**	**0.831**	**0.889**

Table 4: Accuracy and balanced accuracy of the single models

Results: Hybrid-2 Models

	Accuracy			Balanced accuracy		
Hybrid model	Altman	Ohlson	All	Altman	Ohlson	All
RF+XGB	0.946	**0.950**	**0.970**	0.847	0.799	0.866
RF+LGBM	0.943	0.944	0.967	0.807	0.828	0.876
DT+XGB	0.945	0.948	0.966	0.875	0.855	0.890
XGB+LGBM	0.939	0.940	0.966	0.882	0.854	0.896
RF+DT	0.947	0.946	0.963	0.880	0.863	0.897
DT+LGBM	0.940	0.941	0.962	0.840	0.821	0.878
LR+XGB	0.944	0.941	0.959	0.869	0.845	0.891
SVM+XGB	0.933	0.938	0.959	0.877	0.855	**0.910**
LR+RF	**0.953**	0.945	0.957	0.889	0.865	0.896
SVM+RF	0.942	0.943	0.957	0.887	0.854	0.891
LR+LGBM	0.936	0.930	0.950	**0.893**	0.854	0.903
SVM+LGBM	0.928	0.927	0.950	0.897	**0.868**	0.890
LR+DT	0.939	0.935	0.944	0.849	0.828	0.888
SVM+DT	0.930	0.933	0.938	0.841	0.834	0.871
LR+SVM	0.893	0.873	0.893	0.845	0.818	0.893

Table 5: Accuracy and balanced accuracy of the hybrid-2 models (weights 50:50)

Results: Hybrid-2 Models

Split ratio	Model	Balanced accuracy
50:50	SVM + LGBM	0.885
70:30	SVM + XGB	0.889
80:20	SVM + XGB	0.910
90:10	SVM + XGB	**0.929**

Table 6: The best hybrid-2 models

Results: Hybrid-3 Models

Hybrid model	Altman	Ohlson	All
LR+SVM+RF	0.934	0.927	0.938
LR+SVM+DT	0.924	0.919	0.928
LR+SVM+XGB	0.929	0.926	0.941
LR+SVM+LGBM	0.924	0.919	0.938
LR+RF+DT	**0.949**	0.950	0.961
LR+RF+XGB	0.948	**0.951**	0.966
LR+RF+LGBM	0.945	0.942	0.965
LR+DT+XGB	0.947	0.945	0.960
LR+DT+LGBM	0.939	0.937	0.959
LR+XGB+LGBM	0.940	0.938	0.963
SVM+RF+DT	0.947	0.947	0.957
SVM+RF+XGB	0.943	0.948	0.964
SVM+RF+LGBM	0.939	0.939	0.964
SVM+DT+XGB	0.942	0.944	0.957
SVM+DT+LGBM	0.937	0.934	0.954
SVM+XGB+LGBM	0.937	0.935	0.963
RF+DT+XGB	0.948	0.950	0.966
RF+DT+LGBM	0.948	0.944	0.966
RF+XGB+LGBM	0.943	0.945	**0.968**
DT+XGB+LGBM	0.943	0.943	0.967

Table 7: Accuracy of hybrid-3 models (weights 1/3:1/3:1/3)

Results: Hybrid-3 Models

Hybrid model	Altman	Ohlson	All
LR+SVM+RF	0.818	0.815	0.873
LR+SVM+DT	0.870	0.856	0.893
LR+SVM+XGB	0.863	0.849	**0.907**
LR+SVM+LGBM	0.862	0.852	0.897
LR+RF+DT	**0.888**	0.864	0.894
LR+RF+XGB	0.887	0.866	0.905
LR+RF+LGBM	0.882	0.863	0.895
LR+DT+XGB	0.856	0.836	0.894
LR+DT+LGBM	0.871	0.845	0.887
LR+XGB+LGBM	0.874	0.848	0.894
SVM+RF+DT	0.885	0.862	0.886
SVM+RF+XGB	0.878	0.860	0.899
SVM+RF+LGBM	0.887	**0.867**	0.893
SVM+DT+XGB	0.860	0.836	0.884
SVM+DT+LGBM	0.860	0.839	0.890
SVM+XGB+LGBM	0.875	0.851	0.893
RF+DT+XGB	0.868	0.844	0.890
RF+DT+LGBM	0.875	0.842	0.893
RF+XGB+LGBM	0.886	0.846	0.889
DT+XGB+LGBM	0.850	0.826	0.883

Table 8: Balanced accuracy of hybrid-3 models (weights 1/3:1/3:1/3)

Results: Hybrid-3 Models

Split ratio	Model	Acc	Model	Balanced
33:33:33	RF+XGB+LGBM	**0.968**	LR+SVM+XGB	**0.907**
25:25:50	LR+SVM+RF	0.950	LR+SVM+RF	0.898
25:50:25	RF+XGB+LGBM	**0.968**	LR+SVM+XGB	0.904
50:25:25	RF+XGB+LGBM	**0.968**	LR+SVM+XGB	0.903

Table 9: The best hybrid-3 models

Results: Hybrid-4 Models

Hybrid model	Altman	Ohlson	All
LR+SVM+RF+DT	0.909	0.910	0.919
LR+SVM+RF+XGB	0.911	0.911	0.928
LR+SVM+RF+LGBM	0.909	0.905	0.926
LR+SVM+DT+XGB	0.938	0.932	0.951
LR+SVM+DT+LGBM	0.935	0.927	0.947
LR+SVM+XGB+LGBM	0.934	0.928	0.957
LR+RF+DT+XGB	0.934	0.935	0.947
LR+RF+DT+LGBM	0.931	0.924	0.942
LR+RF+XGB+LGBM	0.932	0.928	0.952
LR+DT+XGB+LGBM	**0.940**	**0.941**	**0.962**
SVM+RF+DT+XGB	0.927	0.928	0.950
SVM+RF+DT+LGBM	0.925	0.922	0.944
SVM+RF+XGB+LGBM	0.929	0.926	0.954
SVM+DT+XGB+LGBM	0.934	0.940	0.961
RF+DT+XGB+LGBM	0.935	0.937	0.959

Table 10: Accuracy of the hybrid-4 models

Results: Hybrid-4 Models

Hybrid model	Altman	Ohlson	All
LR+SVM+RF+DT	0.887	0.859	0.901
LR+SVM+RF+XGB	0.876	0.856	0.905
LR+SVM+RF+LGBM	0.879	**0.865**	0.900
LR+SVM+DT+XGB	0.886	0.847	0.897
LR+SVM+DT+LGBM	0.885	0.852	0.891
LR+SVM+XGB+LGBM	0.880	0.857	**0.908**
LR+RF+DT+XGB	0.892	0.856	0.891
LR+RF+DT+LGBM	**0.894**	0.851	0.888
LR+RF+XGB+LGBM	0.887	**0.865**	0.894
LR+DT+XGB+LGBM	0.859	0.835	0.890
SVM+RF+DT+XGB	0.884	0.849	0.904
SVM+RF+DT+LGBM	0.883	0.849	0.894
SVM+RF+XGB+LGBM	0.890	0.860	0.907
SVM+DT+XGB+LGBM	0.851	0.843	0.890
RF+DT+XGB+LGBM	0.860	0.837	0.885

Table 11: Balanced accuracy of the hybrid-4 models

Results: Hybrid-5 and 6 Models

Hybrid model	Altman	Ohlson	All
LR+SVM+RF+DT+XGB	0.926	0.926	0.939
LR+SVM+RF+DT+LGBM	0.924	0.918	0.935
LR+SVM+RF+XGB+LGBM	0.924	0.922	0.941
LR+SVM+DT+XGB+LGBM	**0.938**	**0.935**	**0.956**
LR+RF+DT+XGB+LGBM	0.936	**0.935**	0.952
SVM+RF+DT+XGB+LGBM	0.933	0.934	0.952
LR+SVM+RF+DT+XGB+LGBM	0.886	0.872	0.890

Table 12: Accuracy of hybrid-5 and -6 models

Results: Hybrid-5 and 6 Models

Hybrid model	Altman	Ohlson	All
LR+SVM+RF+DT+XGB	**0.900**	0.851	0.903
LR+SVM+RF+DT+LGBM	0.895	0.855	0.897
LR+SVM+RF+XGB+LGBM	0.895	**0.862**	**0.904**
LR+SVM+DT+XGB+LGBM	0.878	0.852	0.892
LR+RF+DT+XGB+LGBM	0.881	0.848	0.886
SVM+RF+DT+XGB+LGBM	0.883	0.855	0.886
LR+SVM+RF+DT+XGB+LGBM	0.818	0.815	0.873

Table 13: Balanced accuracy of hybrid-5 and-6 models

Results

Model	Model	Accuracy
Single	XGB	**96.2**
Hybrid-2	RF+XGB	**97.0**
Hybrid-3	RF+XGB+LGBM	96.8
Hybrid-4	LR+DT+XGB+LGBM	96.2
Hybrid-5	LR+SVM+DT+XGB+LGBM	95.6
Hybrid-6	LR+SVM+RF+DT+XGB+LGBM	89.0
Model	Model	Balanced Accuracy
Single	LGBM	**88.9**
Hybrid-2	SVM+XGB	**91.0**
Hybrid-3	LR+SVM+XGB	90.7
Hybrid-4	LR+SVM+XGB+LGBM	90.8
Hybrid-5	LR+SVM+RF+XGB+LGBM	90.4
Hybrid-6	LR+SVM+RF+DT+XGB+LGBM	87.3

Table 14: The best single and hybrid models (80:20 split)

Discussion

Model	Training time
LR	0.284
SVM	11.081
RF	1.720
DT	0.329
XGB	0.373
LGBM	0.587
Average	**2.396** seconds

Table 15: Training time of all single models

Discussion

Model	Training time
LGBM	0.587
LR + SVM	56.722
LR + RF	21.778
LR + DT	2.121
LR + XGB	2.945
LR + LGBM	2.859
SVM + RF	99.669
SVM + DT	61.089
SVM + XGB	51.952
SVM + LGBM	51.893
RF + DT	21.600
RF + XGB	23.553
RF + LGBM	22.209
DT + XGB	3.355
DT + LGBM	3.756
XGB + LGBM	3.726
Average	**26.863** seconds

Table 16: Training time of all hybrid-2 models

Conclusion

- Providing empirical evidence that hybrid models can outperform single models in terms of balanced accuracy and overall accuracy.
- Finding the "optimal" hybrid models, specifically hybrid-2 models, for financial distress prediction in emerging economies.
- These models incorporate at least one tree-based method such as XGBoost or LightGBM, demonstrate superior performance.
- *Future work*: Hybrid ML models with neural networks.

Feedback

Minh Nguyen
Email: minhn@msu.edu

Adisa, J. A., Ojo, S. O., Owolawi, P. A., and Pretorius, A. B. (2019). Financial distress prediction: Principle component analysis and artificial neural networks. In *2019 International Multidisciplinary Information Technology and Engineering Conference (IMITEC)*, pages 1–6. IEEE.

Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, 23(4):589–609.

Cao-Van, K., Minh, T. C., Minh, L. G., Be Quyen, T. T., and Tan, H. M. (2024). Soft-voting ensemble model: An efficient learning approach for predictive prostate cancer risk. *Vietnam Journal of Computer Science*.

Huang, Y.-P. and Yen, M.-F. (2019). A new perspective of performance comparison among machine learning algorithms for financial distress prediction. *Applied Soft Computing*, 83:105663.

Lanbouri, Z. and Achchab, S. (2015). A hybrid deep belief network approach for financial distress prediction. In *2015 10th International Conference on Intelligent Systems: Theories and Applications (SITA)*, pages 1–6. IEEE.

Marso, S. and El Merouani, M. (2020). Predicting financial distress using hybrid feedforward neural network with cuckoo search algorithm. *Procedia Computer Science*, 170:1134–1140.

Nguyen, M., Nguyen, B., and Lieu, M. L. (2024). Corporate financial distress prediction in a transition economy. *Journal of Forecasting (forthcoming)*.

Ohlson, J. A. (1980). Financial ratios and the probabilistic prediction of bankruptcy. *Journal of Accounting Research*, 18(1):109–131.